

Analysis Plan: Winter Temperature and Respiratory ED Visits in the Northeast US

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Research question

Are colder winters in Northeast states linked to bigger or earlier flu, RSV, and COVID emergency-department (ED) visit surges?

Hospitals plan staffing and beds around these winter surges. If temperature moves with them, it could be a simple early-warning sign. The plan looks at two things:

1. **Size:** Do colder winters have higher peak ED-visit shares?
2. **Timing:** Do ED visits rise a few weeks after temperatures drop?

Data

Data cover the nine Northeast states (CT, ME, MA, NH, NJ, NY, PA, RI, VT).

Source	What it measures	Level	Seasons
NOAA nClimDiv	Average monthly temperature (°F)	County × month	1895–today
CMU Delphi NSSP	% of ED visits for flu, RSV, COVID	State and county × week	4 (2022–23 to 2025–26)
CMU Delphi FluView	% of doctor visits for flu-like illness (ILI)	State × week	16 (2010–11 to 2025–26)

Note: temperature is monthly, so all weeks in a month share one value. The 2020–21 flu season was almost zero because of COVID measures, so I will flag it and test results with and without it.

Methodology

The main method is exploratory data analysis (EDA) with pandas, seaborn and plotly, plus correlation and one simple linear regression. Results are about association (not causal).

- **Summaries:** `.describe()`, `.groupby().agg()` and `pd.pivot_table()` by state, season and disease.
- **Time series:** line plots of weekly % of visits next to temperature. Use `.rolling()` to smooth, and `.shift()` to line up temperature from 1–4 weeks earlier.
- **Correlation:** `.corr()` between winter temperature and peak %, shown as scatter plots and a correlation heatmap.
- **Simple linear regression:** one line, $\text{peak \%} = a + b \times \text{winter temperature}$, drawn with `sns.regplot()`. The slope b says how much the peak changes per 1 °F warmer.
- **Map:** a plotly map of county winter temperature to show differences across the region.

Analysis steps

1. **(Done) Get the data.** Run the existing NOAA and NSSP scripts.
2. **Clean and merge.** Check for missing values and duplicates, then merge temperature onto each week by state and month.
3. **Summarize seasons.** Use groupby to get each state-season's winter temperature, peak %, and peak week.
4. **Explore.** Plot line charts, histograms and boxplots by state and disease, plus a county temperature map.

5. **Correlate.** Compute correlations and draw scatter plots and a heatmap of temperature vs. peak % and peak week.
6. **Check timing.** Use `.shift()` to compare % of visits with temperature 1–4 weeks earlier, and see which lag has the strongest correlation.
7. **Fit one simple regression** of peak % on winter temperature, first for FluView (16 seasons), then for NSSP (4 seasons).
8. **Write up** the 3–4 clearest charts with a short explanation of what they show.

Limitations

- NSSP has only 4 seasons, so its results are a first look. FluView's 16 seasons are the main test.
- Monthly temperature can miss short cold snaps.
- Other things also drive surges, like holidays, school, vaccines and new COVID variants.